Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 37)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 1 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 1 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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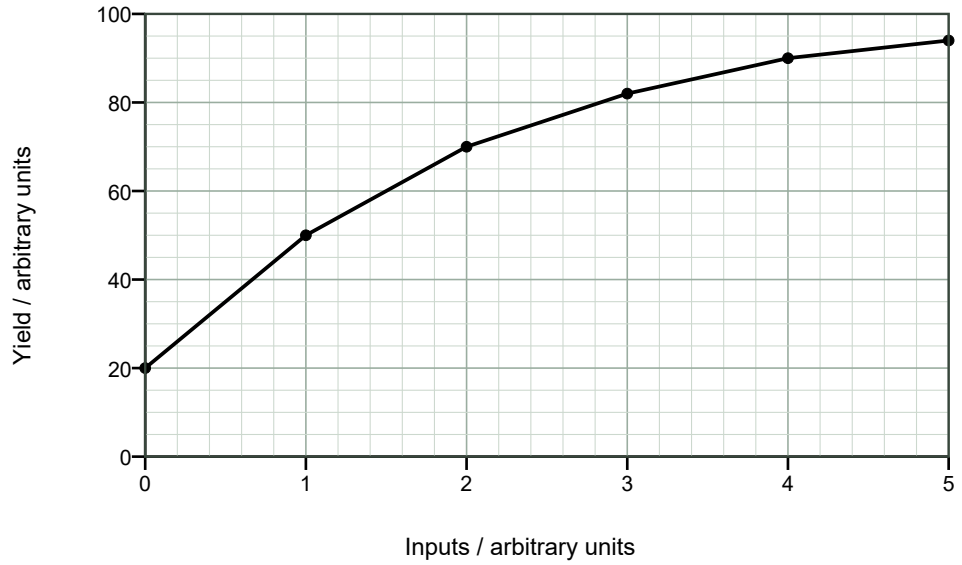
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0 2 Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



0 2 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 2 . 2 Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer _____ units

0 2 . 3 Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

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0 2 . 4 Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

0 2 . 5 Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

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0 3 . 1 Describe how soil texture is determined using a stack of sieves.

[3 marks]

0 3 . 2 Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

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0 3 . 3 Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

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0 3 . 4 Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

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0 4 Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 4 . 1 Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

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0 4 . 2 Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

0 4 . 3 Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

0 4 . 4 Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

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0 5 A salmon farm rears fish in net cages in a sheltered sea loch. Over one production cycle the farm used 2 400 t of feed and produced 2 000 t of salmon.

0 5 . 1 Calculate the food conversion ratio (mass of feed per unit mass of fish produced) for the farm.

[2 marks]

Answer

0 5 . 2 Explain why so much of the world's aquaculture is of carnivorous species such as salmon, and why this reduces the benefit to world food supplies.

[3 marks]

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0 5 . 3 Explain **four** environmental impacts of rearing salmon in net cages in a sea loch.

[4 marks]

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0 5 . 4 Explain **three** ways the environmental impacts of the farm could be reduced.

[3 marks]

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0 5 . 5 Explain why aquaculture cannot simply replace wild capture fisheries.

[3 marks]

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0 6 Table 1 compares organic and inorganic fertilisers.

0 6 . 1 Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

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0 6 . 2 Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

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0 7 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 7 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 7 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 8 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 8 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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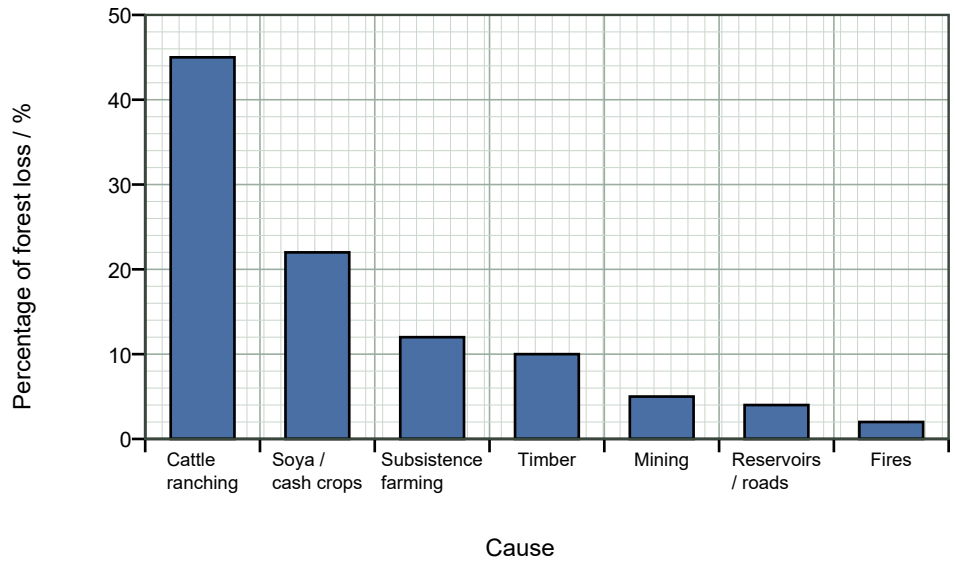
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0 9 Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



0 9 . 1 Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

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0 9 . 2 Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

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0 9 . 3 Describe how each of the following threatens rainforest.

[3 marks]

Mineral extraction .....

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HEP reservoirs .....

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Climate change .....

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0 9 . 4 Describe how a debt-for-nature swap works.

[2 marks]

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1 0 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

1 0 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer ☐ %

1 0 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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1 0 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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1 0 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS